Problem Set 2

Due Saturday, September 19, 11:59 PM.

You may use Mathematica or other computational tools in all problem sets.

1. The wavefunction of a particle of mass m in a one-dimensional space is given to be $\psi(x,t) = Ae^{-\frac{x^2}{4\sigma^2}-\frac{i}{\hbar}Et}$, where A, σ, and E are real constants.

   (a) Write down the corresponding normalized wavefunction. (5 points)

   (b) Determine the probability for finding the particle between $x = 0$ and $x = \infty$. (5 points)

   (c) Determine $\langle\hat{x}\rangle$, $\langle\hat{x}^2\rangle$, and $\Delta x \equiv \sqrt{\langle\hat{x}^2\rangle - \langle\hat{x}\rangle^2}$. (5 points)

   (d) Determine the potential energy of the particle in terms of given parameters. Show that you obtain the standard simple harmonic potential $V(x) = \frac{1}{2}m\omega^2x^2$ for $E = \frac{1}{2}\hbar\omega$ and $\sigma = \sqrt{\frac{\hbar}{2m\omega}}$. (5 points)

2. The wavefunctions and energies for a particle of mass m in a one-dimensional infinite square well with potential $V(x) = 0$ within $0 \le x \le a$ and $V(x) = \infty$ outside are given by $\psi_n(x) = \sqrt{\frac{2}{a}}\sin(\frac{n\pi}{a}x)$ for $0 \le x \le a$, $\psi_n(x) = 0$ outside, and $E_n = \frac{\hbar^2\pi^2n^2}{2ma^2}$, where $n = 1, 2, ...$

   (a) We can rewrite $\psi_n(x) = \sqrt{\frac{2}{a}}\frac{1}{2i}\left(e^{i\frac{n\pi}{a}x} - e^{-i\frac{n\pi}{a}x}\right) = \frac{1}{\sqrt{2}}(\psi_{nR}(x) + \psi_{nL}(x))$, where $\psi_{nR}(x) = -i\sqrt{\frac{1}{a}}e^{i\frac{n\pi}{a}x}$ and $\psi_{nL}(x) = i\sqrt{\frac{1}{a}}e^{-i\frac{n\pi}{a}x}$. Show that $J_n(x) = 0$, $J_{nR}(x) = v\mathcal{P}_{nR}(x)$, and $J_{nL}(x) = -v\mathcal{P}_{nL}(x)$, where $v = \frac{n\pi\hbar}{ma} = \frac{p}{m}$ is velocity, for each of $\psi_n(x)$, $\psi_{nR}(x)$, and $\psi_{nL}(x)$, and point out why these currents are zero, positive, or negative. (5 points)

   (b) Determine $\langle\hat{x}\rangle$, $\langle\hat{x}^2\rangle$, and Δx in state $\psi_n(x)$. Does Δx increase, decrease, or stay the same with increasing n? (5 points)

   (c) Determine $\langle\hat{p}\rangle$, $\langle\hat{p}^2\rangle$, and Δp in state $\psi_n(x)$. Does Δp increase, decrease, or stay the same with increasing n? (5 points)

   (d) Check whether the uncertainty principle, $\Delta x\Delta p \ge \frac{\hbar}{2}$, holds for $\Delta x\Delta p$ in each state $\psi_n(x)$. Does $\Delta x\Delta p$ increase, decrease, or stay the same with increasing n? (5 points)

   (e) What is ΔE in state $\psi_n(x)$? (5 points)

   (f) What is the probability that the particle is found in the left quadrant of the well within $0 \le x \le \frac{a}{4}$ at time $t = 0$? Interpret and discuss the difference between even n and odd n. (5 points)

(g) The particle is in state $\psi(x,0) = c_1\psi_1(x) + c_2\psi_2(x)$ at time $t = 0$, where c_1 and c_2 are real constants.

i. Verify that $\psi(x,t) = c_1\psi_1(x)e^{-\frac{i}{\hbar}E_1 t} + c_2\psi_2(x)e^{-\frac{i}{\hbar}E_2 t}$ at time t. (5 points)

ii. Write down the normalized $\psi(x,0)$ and $\psi(x,t)$. (5 points)

iii. List all possible values of energy that a single measurement could produce at time t and the corresponding probabilities. What is the expectation value of the energy (or average energy) of the particle at time t? (5 points)

iv. Show that the probability density for finding the particle at point x within the well at time t oscillates in time. Determine the frequency and the amplitude of oscillation, and point out how the frequency is related to the energies E_1 and E_2. (5 points)

v. Determine $\langle\hat{x}\rangle$ at time t. (5 points)

vi. Determine $\langle\hat{p}\rangle$ at time t. (5 points)

(h) The particle is in state $\psi(x,0) = \sqrt{\frac{30}{a^5}}\,x(a-x)$ for $0 \le x \le a$ and $\psi(x,0) = 0$ outside the well at time $t = 0$.

i. Make a schematic plot of $\psi(x,0)$. Determine the state $\psi(x,t)$ of the particle at later time t. (5 points)

ii. What are the possible values that a single measurement of energy would produce at time t and the corresponding probabilities. Point out the energy with the highest probability and interpret and discuss the reason or what it means. (5 points)

3. How long did it take you to complete this problem set?